

**Improving Production Planning through Material Flow Analysis: Gas Stove Parts
Manufacturer**

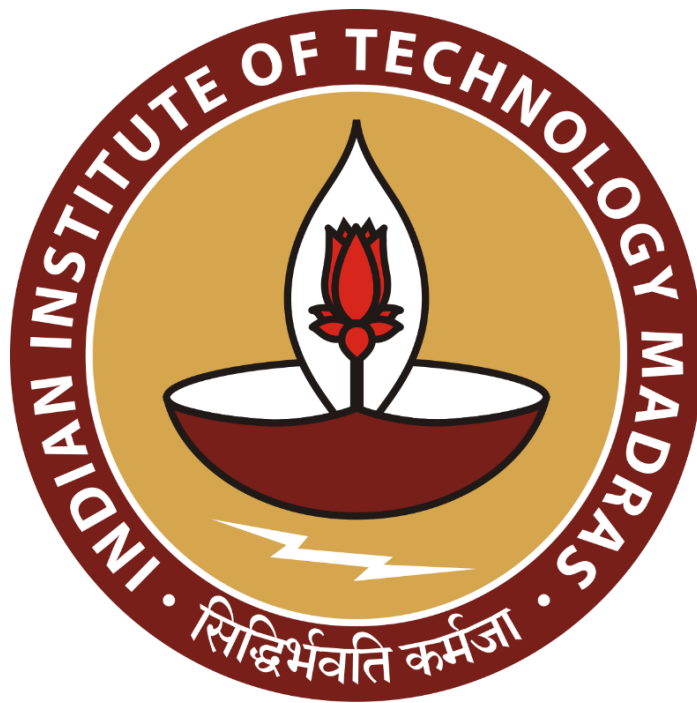
Final Submission Report for the BDM capstone Project

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1. Executive Summary

MetaFab India is an MSME-based B2B manufacturing enterprise in Greater Noida, Uttar Pradesh, producing precision gas stove components including brass burner rings, cast iron pan supports and gas valve assemblies. The business currently relies on manual operational records and faces challenges in production planning due to short order-fulfilment windows, limited production capacity and uncertainty regarding which products and quantities should be prepared before receiving customer orders. The project therefore examines production, procurement, inventory and operational constraints as interconnected elements of production planning.

Primary data were collected through eight datasets covering purchase transactions, sales invoices, production, production capacity, raw-material inventory, finished-goods inventory, worker attendance and machine breakdowns. The 2,529 operational records primarily cover January–June 2026. The analysis employs descriptive statistics, Material Flow Analysis, trend and comparative capacity analysis, coefficient of variation, bottleneck analysis, correlation screening and a three-month weighted moving-average demand signal.

The results indicate substantial procurement variability, with purchase quantity and procurement cost showing coefficients of variation of 73.15% and 77.57%, respectively. Raw-material records show 152,963 kg purchased against 140,786 kg consumed, resulting in a net stock increase of 12,177 kg. Cast Iron Scrap reached 23,593 kg at closing compared with 7,200 kg initially. Finished-goods inventory also increased by 4,961 units, while total production recorded a 4.63% defect rate and 73.1 hours of machine downtime.

The findings suggest that the key opportunity lies in coordinating procurement, inventory buffers, production scheduling and operational constraints. Recommendations therefore focus on procurement review, product-wise make-to-stock/make-to-order decisions, preventive maintenance, workforce planning and improved order-level data capture. No post-implementation business improvement is claimed because the recommendations require operational adoption and subsequent measurement.

2. Detailed Explanation of Analysis Process and Method

2.1 Project Context and Analytical Objectives

The project is grounded in a current production-planning problem rather than a purely technical improvement exercise. Business discussions indicated that the organisation may face difficulty fulfilling customer orders within the required time because production begins after an order is received, the fulfilment window is short and production capacity is limited. Management therefore needs evidence to determine which products and quantities can potentially be prepared in advance

without creating unnecessary inventory. The analysis consequently examines procurement, production, inventory, demand and operational constraints as interconnected components of production planning.

The analysis addresses three project objectives:

- ➔ To evaluate whether raw-material procurement is aligned with actual production requirements.
- ➔ To assess production efficiency by examining production output against customer demand and operational factors.
- ➔ To support production planning by using historical demand patterns to estimate near-term requirements.

These objectives are treated as connected but analytically distinct questions. Each objective determines the datasets, variables and analytical methods used rather than selecting analytical techniques independently of the business problem.

Objective–Method Mapping

Objective	Business Question	Primary Evidence	Analytical Methods
Procurement alignment	Are material purchases and stock levels aligned with production requirements?	Purchase data, raw-material inventory, production data	Descriptive statistics, trend analysis, Material Flow Analysis, inventory coverage
Production efficiency	What measurable operational conditions are associated with production performance?	Production, capacity, attendance, breakdown data	Comparative capacity analysis, coefficient of variation, bottleneck analysis, correlation screening and regression screening
Demand-based planning	What does recent demand imply for near-term production planning?	Sales invoices, finished-goods inventory, production capacity	Demand trend analysis, variability analysis and weighted moving average

This mapping ensures that each analytical method directly contributes to answering a defined business question rather than being included solely for technical complexity. The project therefore prioritises business relevance and interpretability over the number of statistical techniques applied.

2.2 Data Collection and Data Architecture

The project uses primary data obtained directly from MetaFab India. The source records were maintained largely through manual factory registers and were digitised into a structured Microsoft Excel master workbook. Eight datasets were used:

- Purchase transactions
- Sales invoices
- Production records

- Production capacity
- Raw-material inventory
- Finished-goods inventory
- Worker attendance
- Machine-breakdown records

The datasets capture different stages of the business process. Purchase records measure procurement activity, production records capture output and defects, sales invoices provide transaction-level demand and revenue, inventory records measure stock flows, while attendance, capacity and breakdown records capture operational constraints.

The datasets were not blindly merged into one table. Instead, each dataset was retained at its natural level of observation and connected analytically where the variables were compatible. This approach was selected because the datasets represent different processes and frequencies. Purchase and sales transactions were therefore analysed at transaction/month level, inventory data through stock-flow calculations, and production and operational data through daily/product-level analysis.

The eight datasets contain 2,529 rows and 40 columns, with approximately six months of operational coverage. Individual datasets differ in record frequency and exact date coverage, which was considered when performing month-level comparisons and reconciliation.

2.3 Data Quality, Cleaning and Pre-processing

Data cleaning was performed before calculating derived measures or conducting statistical analysis. The purpose was to ensure that differences in formatting, units or data structure did not create artificial patterns or distort business conclusions.

The preprocessing workflow consisted of:

1. Digitisation: Manual factory records were converted into structured worksheets.
2. Source verification: Digitised records were cross-checked against source registers where available.
3. Standardisation: Date, product, material and unit representations were standardised.
4. Numeric validation: Quantity, cost, production, inventory and downtime fields were checked as quantitative variables.
5. Duplicate and missing-value checks: Exact duplicate rows and blank cells were checked across the workbook.
6. Aggregation: Daily and monthly datasets were derived where required for trend and operational analysis.
7. Derived-variable creation: Defect rate, capacity utilisation, inventory coverage, coefficient of variation and net inventory change were calculated.

No blank cells or exact duplicate records were identified in the used ranges at the time of analysis.

An important part of preprocessing was cross-dataset reconciliation. Individual inventory sheets satisfied the stock-flow identity:

$$\text{Closing Stock} = \text{Opening Stock} + \text{Inflow} - \text{Outflow}$$

However, some independent transactional datasets did not reconcile one-to-one. For example, purchase transactions recorded 96,725 kg purchased, whereas the raw-material inventory dataset recorded 152,963 kg. Similarly, differences existed between production/sales transaction records and finished-goods inventory records. These differences were not artificially reconciled or combined. They were retained as a data-governance limitation because the underlying source logs may use different scopes or definitions.

This decision is important because forcing incompatible records into a common total could create false conclusions about procurement, production or inventory.

2.4 Descriptive Statistical Analysis

Descriptive statistics were used as the first analytical layer to establish the scale, central tendency and variability of the major business processes before conducting deeper analysis.

The measures included:

- Mean
- Median
- Standard deviation
- Minimum
- Maximum
- Range
- Coefficient of variation (CV)
- Skewness

The mean was calculated as:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

The sample standard deviation was calculated as:

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

The coefficient of variation was used to compare variability relative to the magnitude of the variable:

$$CV = \frac{s}{\bar{x}} \times 100$$

Skewness was examined to understand whether transaction distributions were symmetric or dominated by unusually large or small observations.

This method was selected because the business problem involves uncertain procurement and demand, making variability as important as average quantities. For example, purchase quantity had a CV of 73.15%, while invoice sales quantity had a CV of 80.06%. Therefore, relying only on average demand or purchase quantities could conceal substantial variation.

2.5 Material Flow Analysis

Material Flow Analysis (MFA) was used to examine whether procurement and production flows were resulting in accumulation or depletion of inventory.

The fundamental material-balance relationship is:

$$Input = Output + Accumulation + Loss$$

For raw materials:

$$Net\ Accumulation = Purchases - Consumption$$

For finished goods:

$$Net\ Accumulation = Production - Sales$$

The inventory balance is therefore:

$$Closing\ Stock = Opening\ Stock + Inflow - Outflow$$

MFA was selected because the raw-material and finished-goods datasets contain the required opening stock, inflow, outflow and closing stock variables. It directly addresses whether procurement and production are becoming disconnected from consumption and customer demand.

For example, Cast Iron Scrap showed purchases of 53,613 kg against consumption of 37,220 kg, producing a net increase of 16,393 kg over the period. This makes it a material-specific procurement/inventory issue rather than evidence that all raw materials are overstocked.

2.6 Procurement and Inventory Analysis

Procurement analysis was performed using transaction-level purchase records to examine:

- Purchasing frequency
- Purchase quantity
- Purchase cost
- Material-wise purchasing behaviour

- Variability

The purchase dataset contains 100 transactions covering five raw-material categories. The difference between mean and median purchase cost, together with a 77.57% coefficient of variation, indicates that procurement includes occasional large purchase events.

Inventory analysis was then performed at material level to determine whether procurement was translating into accumulation or depletion.

Inventory coverage was calculated conceptually as:

$$\text{Days of Cover} = \frac{\text{Closing Inventory}}{\text{Average Daily Consumption}}$$

This measure was used because absolute closing inventory alone does not indicate whether a stock level is high or low. The same quantity may represent very different levels of supply depending on consumption rate.

The combination of procurement variability, material-flow analysis and inventory coverage was selected because the business problem concerns not simply how much material is purchased, but whether purchasing is aligned with actual production requirements.

2.7 Demand and Production Analysis

Customer demand was analysed at both invoice and monthly levels. The analysis examined:

- Sales quantity
- Sales revenue
- Monthly trends
- Product-wise demand
- Weekly demand variability
- Overlap between products appearing in sales and production datasets

This distinction was necessary because the sales dataset contains eight products, whereas production and capacity data cover three manufactured products. Therefore, demand-planning analysis was restricted to the three products for which production and inventory evidence was available, while the remaining five products were treated separately rather than incorrectly assuming they were manufactured internally.

For the three manufactured products, weekly coefficient of variation was used to quantify demand instability. Gas Valve Assembly showed the highest weekly demand variability at approximately 56.5%, while also representing the largest manufactured-product demand.

This analysis directly supports the business question of how much production should potentially be prepared in advance and how much should remain order-driven.

2.8 Production Capacity and Operational Constraint Analysis

Capacity analysis compared observed production with the practical daily capacity specified for each manufactured product.

Capacity utilisation was calculated as:

$$\text{Capacity Utilisation}(\%) = \frac{\text{Average Daily Production}}{\text{Practical Daily Capacity}} \times 100$$

The analysis showed utilisation of approximately 58% across the three products on recorded production days. However, this was not interpreted as automatically available spare capacity because practical output can also be affected by demand, product mix, machine availability, labour availability and raw-material readiness.

This qualification is important to the business problem. The objective is not simply to determine whether capacity exists, but whether existing capacity can be allocated effectively to products with recurring demand and time-sensitive fulfilment requirements.

Machine downtime and worker attendance were therefore incorporated as operational constraints rather than treating production capacity as an isolated variable.

2.9 Correlation and Regression Screening

Pearson correlation was used as an initial screening method to assess whether daily production was linearly associated with major operational variables.

The Pearson correlation coefficient is:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

The variables screened against daily production included:

- Raw-material consumption
- Machine downtime
- Workers present
- Overtime hours
- Invoice sales quantity
- Invoice sales revenue

Correlation was selected as a screening technique rather than a causal model. A correlation indicates the strength and direction of a linear association; it does not establish that one variable causes changes in production.

The strongest observed relationship was between raw-material consumption and daily production, with $r = 0.656$. Machine downtime showed a negative association of $r = -0.352$, while workers present and overtime showed weak relationships.

A multiple linear screening model was additionally used with workers present, overtime hours and machine downtime as explanatory variables:

$$Production = \beta_0 + \beta_1 Workers + \beta_2 Overtime + \beta_3 Downtime + \epsilon$$

The model produced an R^2 of approximately 0.061, meaning that these variables collectively explained only about 6.1% of the observed variation in production. Therefore, the model was not treated as a production forecasting model. Instead, the result supports the conclusion that production planning must consider additional factors such as demand, product mix, material availability, machine constraints and scheduling decisions.

2.10 Short-Term Demand Planning

A forecasting/planning method was required because one of the project objectives is to estimate near-term product requirements.

However, the available dataset contains only approximately six months of observations. Therefore, a complex forecasting model was not considered appropriate because it could create a false impression of predictive precision.

A three-month weighted moving average was selected as a transparent short-term planning signal.

The weighted moving average was calculated as:

$$F_{t+1} = \frac{w_1 D_{t-2} + w_2 D_{t-1} + w_3 D_t}{w_1 + w_2 + w_3}$$

where the weights used were 1, 2 and 3 for April, May and June respectively, giving the greatest weight to the most recent observation.

The method was selected because it:

- Uses recent demand information
- Is transparent to business users
- Does not require a long historical series
- Provides a simple planning benchmark
- Aligns directly with the question of what quantity may potentially be prepared before customer orders are received

The resulting planning signals were approximately 1,170 Brass Burner Rings, 2,562 Gas Valve Assemblies and 492 Cast Iron Pan Supports. These values are treated as planning signals rather than

fixed production targets because the limited historical window and demand variability do not justify treating the estimates as precise forecasts.

2.11 Integration of Analytical Findings

The methods were not interpreted independently. The final analytical interpretation combines the evidence from procurement, inventory, demand and operational analysis.

The overall analytical logic is:

Business Problem → Relevant Data → Data Validation → Descriptive Analysis → Process-Specific Analysis → Interpretation → Recommendation

For MetaFab India, this integration is particularly important because the analysis does not indicate that a single factor explains the production-planning challenge. Procurement variability, inventory accumulation, demand variability, capacity utilisation and operational constraints must be considered together.

Therefore, the analytical process ultimately evaluates how available material, production capacity and finished-goods inventory can be aligned with observed customer demand while reducing the risk of unnecessary pre-production.

3. Results and Findings

3.1 Overall Operational Profile

KPI	Observed result
Dataset records	2,529 rows across 8 datasets
Purchase transactions	100
Sales invoices	315
Production records	464
Production days	180
Good units produced	185,173
Defective units	8,992
Invoice sales revenue	₹4.831 million
Raw-material purchases in inventory system	152,963 kg
Raw-material consumption	140,786 kg
Finished goods produced in inventory system	175,775 units
Finished goods sold in inventory system	170,814 units

KPI	Observed result
Machine downtime	73.1 hours
Worker attendance	88.2% of planned presence

The overall profile shows a manufacturing system with stable ongoing output but substantial variability in the inputs and supporting processes. The business is not simply facing a shortage of capacity. Instead, the data points toward coordination: procurement is variable, inventory absorbs some mismatches, production operates below practical capacity, and operational reliability varies across machines and workforce availability.

Gas Valve Assembly contributes the largest production volume at 94,135 good units, followed by Brass Burner Ring at 63,041 and Cast Iron Pan Support at 27,997. The production mix is therefore concentrated in Gas Valve Assembly, which is also the largest manufactured-product revenue contributor in the invoice data. This makes it an especially important product for planning decisions.

3.2 Procurement Pattern

Purchase transactions show a non-uniform procurement pattern. Total transaction purchases are 96,725 kg at a total recorded cost of ₹14.941 million. Purchase quantity has a median of 772 kg against a mean of 967 kg and a standard deviation of 707.55 kg, confirming a right-skewed distribution with occasional large purchase events.

Material	Purchased kg	Qty share	Procurement cost ₹	Cost share	Weighted ₹/kg
RM001 Brass Ingots	8150.0	8.4	4704032.0	31.5	577.18
RM002 Cast Iron Scrap	40621.0	42.0	2028435.0	13.6	49.94
RM003 ADC12 Aluminium	19068.0	19.7	4635266.0	31.0	243.09
RM004 SS304 Sheet	10449.0	10.8	2233717.0	14.9	213.77
RM005 CRCA Sheet	18437.0	19.1	1339908.0	9.0	72.67

Two materials dominate procurement differently. Cast Iron Scrap represents approximately 42.0% of purchased quantity but only 13.6% of procurement cost because its unit price is comparatively low. Brass Ingots account for only 8.4% of purchased quantity but 31.5% of procurement cost, making small quantity decisions financially important. ADC12 Aluminium similarly accounts for 31.0% of procurement cost. Procurement review should therefore consider both quantity and monetary exposure, not quantity alone.

The inventory-system view shows positive net material accumulation in January (+2,696 kg), February (+7,336 kg), April (+6,818 kg) and May (+2,674 kg), while March and June consume more than they purchase. February shows the largest positive monthly gap between inventory purchases and consumption, suggesting that procurement is being used to build a buffer. The final planning system should explicitly distinguish intentional safety stock from unplanned accumulation.

3.3 Raw-Material Inventory

Cast Iron Scrap is the clearest inventory imbalance in the uploaded records. Its closing stock of 23,593 kg is more than three times the opening stock of 7,200 kg, while total purchases exceed consumption by 16,393 kg. At the material's observed average daily consumption rate, the closing balance corresponds to approximately 114.7 days of coverage. This does not by itself prove that the stock is excessive—supplier lead time, minimum order quantity and future orders may justify buffers—but it is sufficiently large to warrant an explicit procurement review.

Brass Ingots present the opposite risk. Closing stock is 1,972 kg against opening stock of 4,800 kg, and purchases are lower than consumption over the period. The resulting approximate coverage is 13.3 days at the observed average consumption rate. This material therefore warrants a different management policy from Cast Iron Scrap: further reduction in stock could create production-risk exposure if supplier lead time is significant.

ADC12 Aluminium, CRCA Steel Sheet and SS304 Sheet show moderate declines in closing stock and approximately 12–28 days of observed consumption coverage. The data supports a differentiated material policy rather than a single reorder rule for all raw materials.

Finished-goods inventory in its own operational ledger rises from 500 units at the opening of January to 5,461 units at the end of June. In absolute terms, the net build is 4,961 units, equal to 2.82% of the 175,775 units produced in that ledger. The modest cumulative build, combined with positive net additions in every month, indicates a persistent but not explosive inventory accumulation pattern.

3.4 Production Performance

The production dataset contains 464 product-day records across 180 production days. Total good output is 185,173 units, average output per production record is 399.08 units, and the overall defect rate is 4.63%. Daily aggregate production averages 1,028.74 good units with a coefficient of variation of 32.07%, showing meaningful day-to-day variability once product records are consolidated at the day level.

Gas Valve Assembly has the highest production volume and the highest product-level defect rate among the three manufactured products at 4.79%, followed by Cast Iron Pan Support at 4.58% and

Brass Burner Ring at 4.42%. The differences are not large enough to justify treating any product as a standalone quality crisis, but the product-level view indicates where further process-level investigation should begin.

Raw-material consumption is positively correlated with daily production at $r = 0.656$, which is expected because higher production activity generally requires more material. This relationship is useful for planning material readiness, but it is not a causal model: both variables are generated by the same production process.

The demand signal shows that recent manufactured-product demand is materially below practical daily capacity. The implication is that increasing capacity is unlikely to be the first action. A stronger planning opportunity is to choose the right product mix and decide where limited pre-production is justified by recurring demand and short fulfilment requirements.

3.5 Customer Demand and Sales

The sales invoice dataset contains 315 invoices covering 147 unique dates and 15 customers. Total invoice revenue is ₹4.831 million. Revenue rises from ₹0.765 million in January to ₹0.940 million in June, while invoice quantity rises from 11,040 units to 13,574 units over the same months. This indicates an overall positive sales trend during the study window despite month-to-month variation.

Invoice Product	Units sold	Revenue ₹	Revenue share
Gas Valve Assembly	15357	1521979.0	31.5
Aluminium Mixing Tube	13023	926000.0	19.2
Brass Injector	6741	467000.0	9.7
Stainless Steel Drip Tray	6289	454000.0	9.4
Brass Burner Ring	5216	442529.0	9.2
Cast Iron Pan Support	3067	391915.0	8.1
Gas Pipe Connector	1935	176000.0	3.6
Brass Burner Head	1245	45000.0	0.9

Gas Valve Assembly is the largest revenue contributor in the invoice data at ₹1.522 million, followed by Aluminium Mixing Tube at ₹0.926 million. For the three products that appear in both production and capacity records, Gas Valve Assembly contributes 64.6% of their invoice revenue, compared with 18.8% for Brass Burner Ring and 16.6% for Cast Iron Pan Support. This concentration makes Gas Valve Assembly a key candidate for a carefully controlled production-planning policy.

A critical scope issue is visible here: five products appear in sales invoices but have no production or capacity records. The final report therefore does not assume that all invoice products are manufactured internally. These five products are excluded from product-level production-capacity and manufacturing-mix conclusions unless the business confirms that they are internally produced.

3.6 Capacity Utilisation

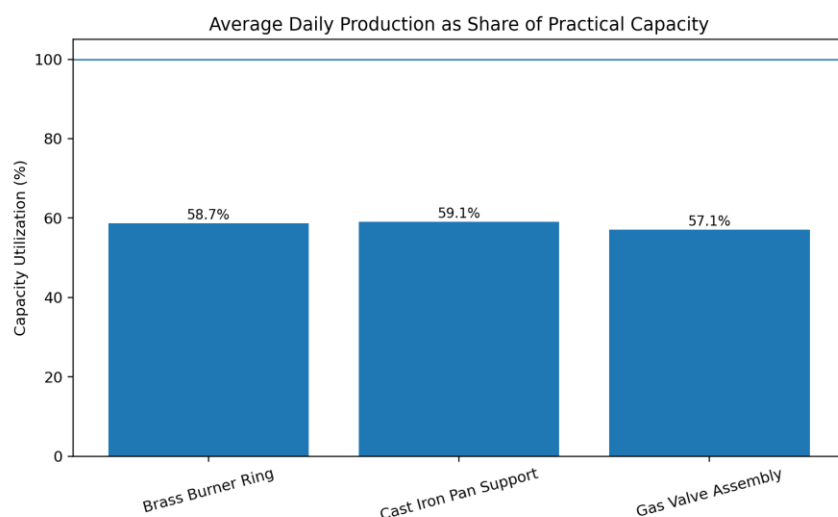


Figure 3.3. Practical capacity versus average daily production

Product	Practical cap/day	Avg. output / recorded day	Utilisation	Gap / day
Brass Burner Ring	680	398.99	58.68%	281.01
Cast Iron Pan Support	320	189.17	59.12%	130.83
Gas Valve Assembly	1043	595.79	57.12%	447.21
Overall	2043	1,183.95	57.95%	859.05

The three manufactured products operate at roughly 57–59% of practical capacity when average production is calculated over days on which the product has recorded production. The largest absolute daily gap is for Gas Valve Assembly at roughly 447 units per recorded production day, while Cast Iron Pan Support has the highest percentage utilisation at 59.1%.

The key management implication is that the business appears to have some available production headroom, but that headroom cannot automatically be treated as free capacity. Production is constrained by product mix and by the need to respond to orders within short fulfilment windows. Capacity should therefore be reserved or allocated based on expected demand and confirmed material and machine availability.

An important methodological note is that the existing capacity code averages only records where a product appears in the production dataset. If a product is not scheduled on a particular day, that day is absent from the average. A stricter utilisation measure using every calendar production day would be lower. The final report therefore treats the approximately 58% figure as "utilisation on recorded production days," not as a plant-wide utilisation rate.

3.7 Machine and Workforce Constraints

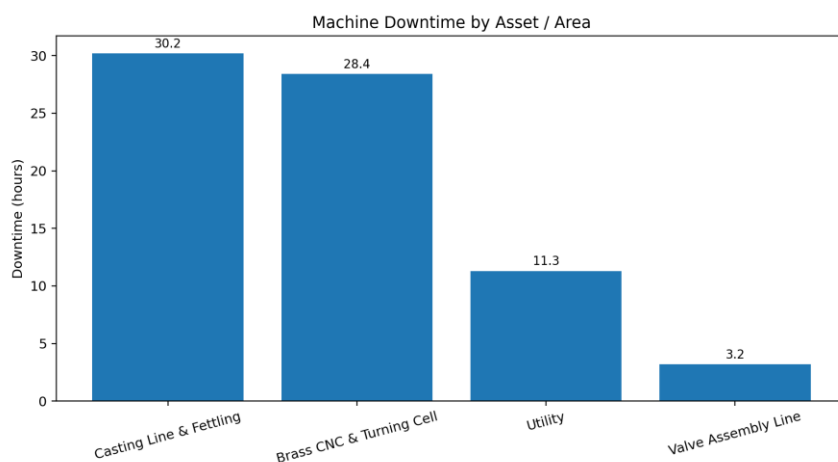


Figure 3.4. Machine downtime by asset / operating area

The machine breakdown log contains 18 events totalling 73.1 hours of downtime, with an average of 4.06 hours per breakdown. The Casting Line & Fettling contributes 30.2 hours (41.3% of total downtime), while the Brass CNC & Turning Cell contributes 28.4 hours (38.9%). Together these two areas account for approximately 80.2% of downtime and should therefore receive priority in preventive-maintenance planning.

Breakdown reason	Events	Downtime hours	Share
Molten Metal Leakage	3	12.7	17.4
Servo Alarm	2	12.7	17.4
Sand Mold Jam	3	12.4	17.0
Power Failure	4	11.3	15.5
Coolant Pump Failure	2	9.0	12.3
Spindle Fault	2	6.7	9.2
Hydraulic Failure	1	5.1	7.0
Conveyor Jam	1	3.2	4.4

Repeated causes are more informative than the total event count. Molten metal leakage, servo alarms and sand-mould jams each account for roughly 12–13 hours of downtime, while power failure contributes 11.3 hours. Preventive maintenance should therefore focus on the recurring causes observed in the breakdown register rather than treating every machine equally.

Average worker attendance is 88.2% of planned presence, with 2.46 hours of overtime per day on average. However, daily production has almost no linear correlation with worker presence ($r = 0.05$) and only a weak relationship with overtime ($r = 0.18$). This prevents a simplistic conclusion that higher headcount alone will solve production variability. The more defensible interpretation is that workforce availability becomes important in combination with product mix, machine status and scheduling.

3.8 Integrated Findings

The project findings converge on five interrelated observations.

1. Procurement is highly variable. Transaction purchase quantity and cost both have coefficients of variation above 70%, indicating that purchasing decisions include occasional large events and should be monitored against inventory requirements rather than average quantity alone.
2. Raw-material inventory is unevenly balanced. Cast Iron Scrap shows a large accumulation and very high observed coverage, while Brass Ingots show a substantial reduction in closing stock. The business therefore needs material-specific inventory policies.
3. Finished-goods inventory rises gradually. The inventory ledger records 4,961 more finished units at the end of June than at the beginning of January. This is only 2.82% of the six-month ledger production, but the direction is consistently positive and should be monitored because the business already faces a trade-off between short fulfilment windows and carrying stock.
4. Production has available headroom but does not automatically justify capacity expansion. Product-level utilisation is around 57–59% on recorded production days, indicating that the key question is allocation and scheduling, not simply installed capacity.
5. Operational reliability needs focused intervention. The Casting Line & Fettling and Brass CNC & Turning Cell together account for about 80% of recorded machine downtime, while workforce attendance alone explains little of daily production variation.

The overall evidence supports an integrated production-planning approach in which demand expectations, material coverage, capacity headroom and operational reliability are reviewed together before a production order is released. This is consistent with the project's original objective of moving from manually maintained records to evidence-based decision support.

Finding	Business implication	Priority
Very high procurement variability	Review purchases against material-specific stock coverage and future requirements	Immediate
Cast Iron Scrap accumulation	Potential working-capital and storage exposure; procurement needs review	Immediate
Brass Ingots stock decline	Potential future production risk if replenishment lead time is long	Immediate
57–59% utilisation on recorded production days	Available headroom can support selective pre-production without new capacity	Medium
80% of downtime concentrated in two operating areas	Maintenance effort can be prioritised where impact is greatest	Immediate
Invoice demand variable and product-concentrated	Planning should use product-specific demand rather than plant-wide averages	Medium

4. Interpretation of Results and Recommendations

4.1 Interpretation of Results

The central interpretation is that MetaFab's production-planning problem is not caused by one isolated operational weakness. The business has enough practical production capacity on recorded production days, but the timing of orders, material availability, inventory accumulation and machine reliability create coordination challenges. The data therefore supports a planning-control problem rather than a simple capacity-shortage problem.

The most important trade-off is between responsiveness and inventory. Producing only after receiving an order reduces the risk of unnecessary finished-goods stock but increases the risk of late fulfilment when the production window is short. Pre-producing too aggressively can reduce lead time but consumes working capital and may create inventory that moves slowly. The appropriate decision is therefore product-specific and evidence-based.

The project also demonstrates why business interpretation must go beyond charts. For example, high procurement variability cannot be labelled simply as "good" or "bad". It must be interpreted together with inventory coverage, consumption and the expected production plan. Likewise, lower machine downtime would be desirable, but the data does not justify claiming that downtime is the sole cause of

production variability because the daily production model using workers, overtime and downtime explains only 6.1% of total variation.

4.2 Recommendations

Recommendation	Action	Review Horizon	Expected Business Benefit
1. Material-specific procurement review	Create a monthly review for each raw material using consumption, closing stock, recent procurement and supplier lead time. Immediately review further Cast Iron Scrap purchases because the current ledger shows 23,593 kg closing stock and ~114.7 days of observed consumption coverage.	Weekly stock review; monthly procurement reset	Reduce unplanned accumulation while avoiding shortages.
2. Separate make-to-stock and make-to-order products	Use recent demand frequency and variability to identify products eligible for controlled pre-production. Gas Valve Assembly should receive priority analysis because it is the largest manufactured-product demand contributor but also has the highest weekly demand variability.	Pilot for 8–12 weeks	Lower fulfilment pressure without blanket inventory build-up.
3. Use a transparent short-term demand signal	Use a three-month weighted demand signal as a planning input, not as an automatic production target. Recalculate monthly as new orders arrive.	Monthly	Improve near-term scheduling consistency.
4. Prioritise maintenance on the two largest downtime areas	Focus preventive-maintenance effort on Casting Line & Fettling and Brass CNC & Turning Cell, which together contribute ~80.2% of recorded downtime. Review repeated failure	Start immediately; review monthly	Reduce avoidable production interruption.

Recommendation	Action	Review Horizon	Expected Business Benefit
	causes such as molten-metal leakage, servo alarms and sand-mould jams.		
5. Use capacity headroom selectively	Do not invest in additional capacity based on current data. Instead, allocate available headroom to products with recurring demand, confirmed material availability and short fulfilment windows.	Quarterly capacity review	Improve utilisation before capital expansion.
6. Improve order-level data capture	Capture order-received date, promised dispatch date, actual dispatch date, order quantity and product for every customer order.	Implement within one month	Make fulfilment delay and service level directly measurable.
7. Build a product-material mapping	Maintain a bill-of-materials mapping from each manufactured product to raw materials and standard material quantity.	Implement within one month	Enable product-level material-flow and purchase planning.

The recommendations are deliberately conservative. They use the current data to identify where attention should be focused, while avoiding unsupported assumptions about supplier lead times, customer service targets or profitability. Any operating threshold should be calibrated with the business after the missing order-level and bill-of-material information is captured.

4.3 Recommended Production-Planning Framework

The proposed framework converts the analytical results into a repeatable management routine. It is designed to be practical within the organisation's existing environment and can initially be implemented in Excel before being automated in Python or another reporting platform.

Step	Planning Stage	Decision Input
1	Demand signal	Recent product demand, order pipeline and weighted short-term demand estimate
2	Inventory position	Raw-material closing stock, finished-goods stock and consumption coverage

Step	Planning Stage	Decision Input
3	Capacity check	Practical capacity versus planned output and current utilisation
4	Constraint check	Machine availability, downtime history and workforce availability
5	Planning decision	Make-to-stock, make-to-order or material-readiness decision by product
6	Execution monitoring	Production achieved, defects, dispatch performance and inventory change
7	Feedback loop	Monthly recalibration of demand, stock policies and maintenance priorities

A concise decision rule is: do not pre-produce solely because capacity is available; pre-produce only when recent demand is sufficiently recurring, the product has a short fulfilment requirement, the required material is available, and the resulting stock remains within an explicitly controlled buffer. Otherwise, retain an order-driven strategy while improving material readiness and capacity availability.

4.4 Implementation and Expected Impact

Horizon	Implementation Action
0–30 days	Reconcile data definitions; add order and dispatch timestamps; confirm bill-of-material mappings; review Cast Iron Scrap and Brass Ingots stock positions.
31–60 days	Pilot product-level planning rules and maintenance priorities; use the weighted demand signal as a planning input; monitor stock and fulfilment KPIs.
61–90 days	Evaluate pilot outcomes; recalibrate stock buffers and production schedules; formalise monthly production-planning review.
Quarterly	Review capacity utilisation, demand variability, inventory coverage, machine downtime, defect rate and fulfilment performance.

The principal expected impact is improved decision quality rather than an unsupported promise of a specific revenue increase. Better alignment should reduce the risk of simultaneous excess inventory and delayed fulfilment, improve use of existing production capacity, focus maintenance spending on the most significant downtime sources and give management a repeatable quantitative planning routine.

4.5 Limitations

- The study covers approximately six months, so long-term seasonality cannot be established reliably.
- The machine-breakdown dataset contains only 18 events; downtime-related relationships should therefore be treated as directional.
- The operational ledgers have different scopes and do not reconcile one-to-one. The report therefore avoids forcing unlike measures into a single mass balance.
- The sales invoice file contains eight products, while production and capacity records contain three. Manufacturing conclusions are restricted to the three common products unless the business confirms otherwise.
- Order-received, promised-dispatch and actual-dispatch timestamps, as well as a verified product-to-material bill of materials, are not available. Fulfilment performance and complete product-level material requirements therefore remain areas for future data capture. The weighted moving-average demand signal should also be recalculated as new observations become available.

5. Supporting Links and Conclusion

Dataset Link	Data Analysis, Approach, Outputs and Outcomes
Exploratory Data Analysis & Visualizations	Metafab India: Experience in the Company

5.1 Final Conclusion

This project converts MetaFab India's manually maintained operational records into a quantitative production-planning view. The evidence shows practical capacity headroom on recorded production days, while planning efficiency is constrained by procurement variability, uneven raw-material inventory, positive finished-goods accumulation, variable demand and concentrated machine downtime. The strongest immediate opportunity is coordinated planning rather than additional capacity investment.

A data-driven planning routine should review demand, inventory, material readiness, capacity and operational constraints together before releasing production decisions. Product-specific pre-production can be considered where recurring demand and short fulfilment windows justify it, while material-specific procurement controls should prevent unnecessary accumulation. Maintenance priorities should focus on the operating areas responsible for the largest share of downtime, and future data capture should add order and bill-of-material information so that fulfilment and product-level material requirements can be measured directly. The framework is intended as decision support that converts fragmented records into a repeatable planning process.